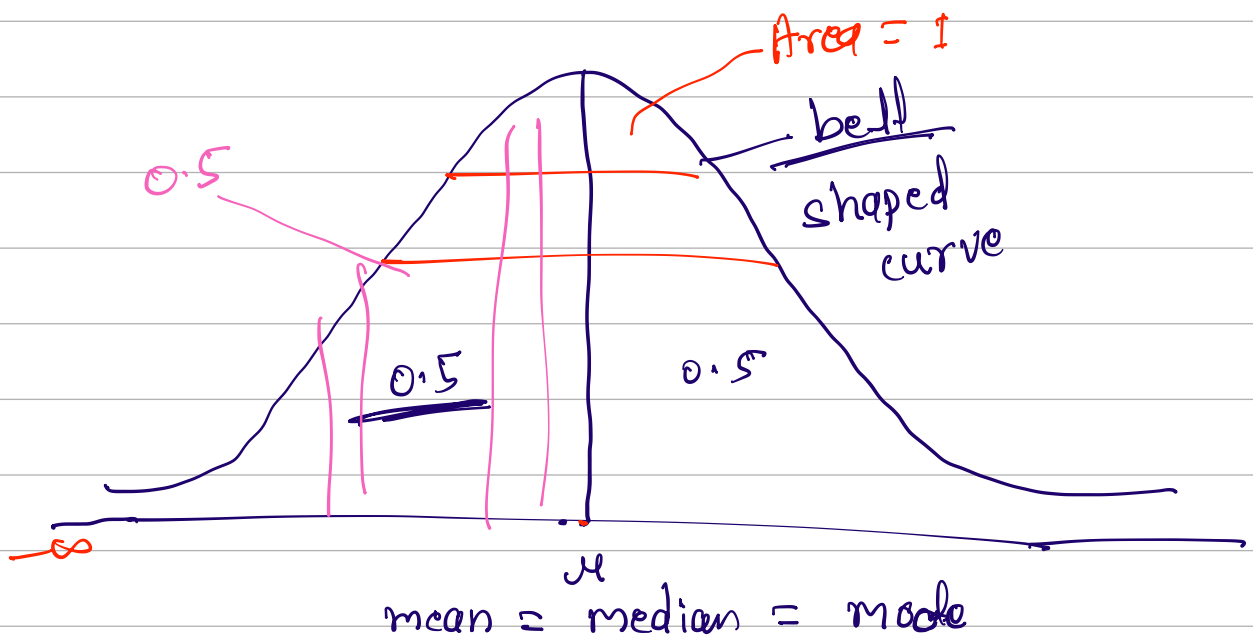


Normal distributⁿ: A random variable X follow a normal distributⁿ with mean μ and standard deviation σ if its probability density functⁿ (PDF) is given by

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty \quad \text{--- (1)}$$



$X \sim N(\mu, \sigma^2)$ → Way to denote normal distributⁿ.

special case → if $\mu = 0$ and $\sigma = 1$, then it is called standard normal distributⁿ.

e.g. A professor takes an exam, and the score is normally distributed with mean $\mu = 70$ and S.D. = $\sigma = 10$. Find the probability that a score is above 85.

solutⁿ:

$$X = \text{score of exam} \sim N(70, 10^2)$$

$$\begin{aligned}
 \underline{P(X > 85)} &= ? \\
 &= \underline{1 - P(X \leq 85)} \\
 &\quad \text{CDF}(85) \rightarrow
 \end{aligned}$$

CDF of the normal distributⁿ:

$$P(X \leq \underline{t}) = \text{cdf}(t) = \int_{-\infty}^t \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Let $t > \mu$

$$\text{CDF}(t) = \int_{-\infty}^{\mu} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx + \int_{\mu}^t \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \underline{0.5} + \int_{\mu}^t \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$\text{Let } \underline{Z = \frac{x-\mu}{\sqrt{2}\sigma}} \Rightarrow dz = \frac{dx}{\sqrt{2}\sigma}$$

limit $\rightarrow$ lower $\rightarrow Z = 0$

$\rightarrow$ upper $\rightarrow Z = \frac{t-\mu}{\sqrt{2}\sigma}$

$$\text{CDF}(t) = 0.5 + \int_{Z=0}^{\frac{t-\mu}{\sqrt{2}\sigma}} \frac{1}{\sqrt{\pi}} e^{-z^2} dz$$

not possible analytically

$\downarrow$
we will use numerical approach.

$$\boxed{\text{erf}(\underline{y}) = \frac{2}{\sqrt{\pi}} \int_0^{\underline{y}} e^{-t^2} dt} \quad \text{(error funtⁿ)}$$

numerically integrate e^{-t^2}

From equatⁿ ② CDF become

$$CDF(t) = 0.5 + \frac{1}{2} \int_{z=0}^{\frac{t-\mu}{\sqrt{2}\sigma}} \frac{2}{\sqrt{\pi}} e^{-z^2} dz$$

$$\boxed{CDF(t) = 0.5 + 0.5 \times \text{erf}\left(\frac{t-\mu}{\sqrt{2}\sigma}\right)} \quad \text{③}$$

*

Exponential distributⁿ:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & , \quad x \geq 0, \quad \boxed{\lambda > 0} \\ 0 & , \quad x < 0 \end{cases}$$

$$\begin{aligned} CDF(t) &= \int_0^t \lambda e^{-\lambda x} dx \\ &= \left(\frac{\lambda e^{-\lambda x}}{-\lambda} \right)_0^t = -e^{-\lambda t} + 1 \\ &= 1 - e^{-\lambda t} \end{aligned}$$